

Project Report: VistaBank ATM System

1. Introduction

The **VistaBank ATM System** is a console-based C program that simulates an ATM and an admin management system. It allows users to perform banking operations such as checking balance, withdrawing cash, and changing passwords, while administrators can create or delete accounts, deposit cash, and view transaction history.

2. Objectives

- Provide a simple, efficient, and secure ATM experience.
- Allow administrators to manage accounts effectively.
- Ensure secure transactions and password protection.

3. Features and Functionalities

3.1 Admin Portal

Admins have access to:

- **Create an Account:** Adds a new user with an ID and password.
- **Delete an Account:** Removes an existing account.
- **Deposit Cash:** Adds funds to a user's account.
- **View All Accounts:** Lists all registered accounts.
- **Change Account Password:** Allows admins to reset user passwords.
- **View All Transactions:** Displays transaction history for all users.

Security Features:

- Admin authentication using **Admin ID and Password**.
- Admin IDs must be **4-digit numeric values**.
- Password masking for input security.

3.2 ATM Portal

Users can:

- **Check Balance:** Displays the current account balance.
- **Withdraw Cash:** Allows users to withdraw money if sufficient balance exists.
- **Change Password:** Enables password updates for security.

Security Features:

- User authentication using **User ID and Password**.
- Only numeric 4-digit IDs allowed.
- Password input is masked.

3.3 Data Management

- **Accounts are stored in an array** with a limit of 50 accounts.
- **Transaction history** is recorded for up to 100 transactions per account.
- **Account persistence** through file handling (loading and saving accounts).

4. Code Structure and Implementation

4.1 Data Structures

- **Struct Account:**
 - Stores ID, password, balance, and transaction history.
 - Maintains the count of transactions per account.

4.2 Functions Overview

Function	Description
main()	Loads accounts and displays the main menu.
displayMainMenu()	Shows the options for Admin and ATM portals.
adminPortal()	Handles admin authentication and admin operations.
atmPortal()	Handles user authentication and ATM operations.
createAccount()	Allows admins to add new accounts.
deleteAccount()	Enables admins to remove accounts.
depositCash()	Admin can deposit cash into user accounts.
viewAllAccounts()	Displays all accounts.
changeAccountPassword()	Allows admin to reset passwords.
viewAllTransactions()	Displays all transactions for all users.
checkBalance()	Shows the balance of a user.

Function	Description
withdrawCash()	Allows users to withdraw cash.
changeUserPassword()	Enables users to update their password.
maskPassword()	Hides password input for security.

5. Error Handling & Security Measures

1. Input Validation:

- **Admin & User IDs must be numeric and exactly 4 digits.**
- **Duplicate account prevention** when creating new accounts.
- **Validations before deleting an account.**

2. Password Security:

- **Masked password entry** for security.
- **Strict password format (4-digit numeric values only).**

3. Data Integrity:

- **Transaction history maintained per account.**
- **Prevention of unauthorized withdrawals** by checking balance.

6. Limitations & Future Improvements

Current Limitations

- No real-time database for persistent storage (currently uses arrays).
- Transaction history is stored in memory and may be lost upon exit.
- Admins cannot edit account details except passwords.

Potential Improvements

- Implement **file handling** to save and load accounts persistently.
- Introduce **dynamic memory allocation** instead of fixed array size.
- Add **Graphical User Interface (GUI)** for better usability.
- Implement **encryption** for storing passwords securely.

7. Conclusion

The **VistaBank ATM System** is a well-structured **C program** designed for secure banking transactions. It provides **basic ATM functionalities** along with an **admin management portal**. With minor improvements like **file handling** and **database integration**, this system can be expanded for real-world applications.